Bachelor Project

Czech Technical University in Prague

F3

Faculty of Electrical Engineering
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Design of a platform for 2D LiDaR

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Field of study: Electronics and Communications
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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 13. November 2025

Abstract

There are currently many advanced LiDAR sensors available for 3D spatial scanning [*insert citation]. However, most of these sensors are too expensive for the average user. The most affordable commercial option is the Unitree L1, which costs $ 250 [*insert citation]. The aim of this miniproject is to design a platform for a 360° LiDAR sensor, which will be utilized in a subsequent bachelor's thesis to visualize a 3D image of the scanned environment.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

Supervisor: Ing. Pavel Puričer, Ph.D.

Abstrakt

V současnosti existuje spousta pokročilých lidar-senzorů určených pro 3D snímání prostoru [*insert citation*]. Většina z nich je však cenově nedostupná pro běžného uživatele. Nejvíc dostupným komerčním řešením je Unitree L1, který stojí $ 250 [*insert citation*]. Cílem toho miniprojektu je navrhnout pro 360° lidar-senzor platformu a použít ji v navazující bakalářské práci pro vizualizaci 3D-obrazu naskenovaného prostoru.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDaR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localization and Mapping

Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [1; 2, Ch. 4-6]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the cheapest commercial 3D LiDAR sensor, which was found during the research for this miniproject, costs $249 [3]. The primary motivation behind this mini-project (and the subsequent bachelor's thesis) is to develop a cost-effective platform for a 360° LiDAR sensor that can be utilised for 3D spatial scanning and visualisation of the surrounding environment. A subsequent bachelor's thesis will then focus on the Digital Signal Processing (DSP) techniques for the effective processing of the acquired point cloud data and proper utilisation of open-source visualisation tools (such as Python's Open3D library [4]) to recreate a 3D scene.

Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms which the author has stumbled upon during the research[too informal?], analyzing their pros and cons. After that, the proposed design will be presented, along with the chosen design decisions and their justifications.

2.1 Existing DIY Solutions for Environmental Mapping

2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilize a single-point LiDAR module mounted on a platform which is able to rotate and tilt. The implementation could be found in this project: [5].

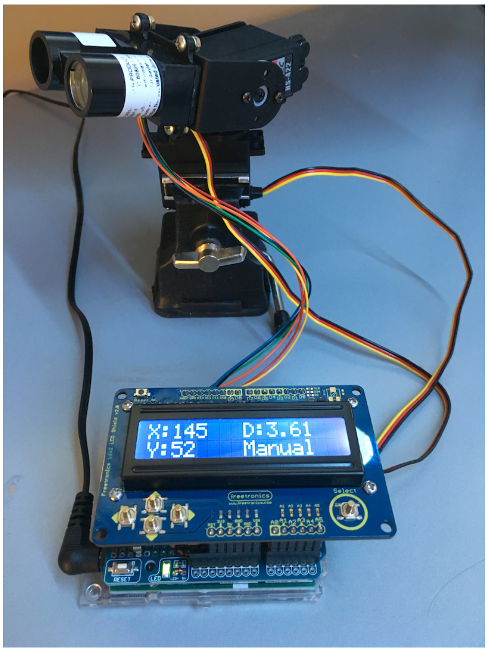


Figure 2.1: Implementation in project [5] using a single-point LiDAR

Pros

+ Single-point LiDARs are the most cost-effective among all existing types
+ Simple design

Cons

– Slow data acquisition relative to methods leveraging 2D LiDAR's speed
– Movement of lidar may introduce innacuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project: [6]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360°. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

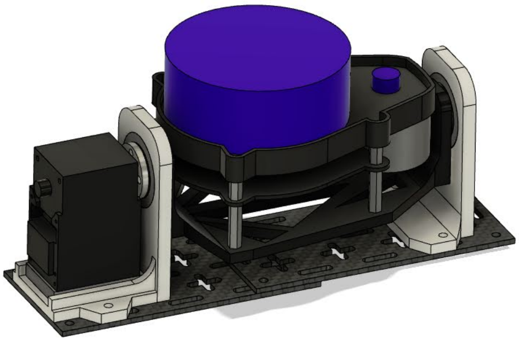


Figure 2.2: Implementation in project [6] using a single-point LiDAR and a mirror

Pros

+ Cost-effectiveness as in previous implementation
+ Simple design
+ Much easier to correct the distance measurement, since the LiDAR is stationary and the distance from it to the mirror is known.

Cons

– Slow data acquisition relative to methods leveraging 2D LiDAR's speed
– Mirror's quality may affect the accuracy of distance measurements
– Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilizes a 2D LiDAR sensor mounted on a platform that can tilt the sensor side-to-side, as demonstrated in this project: [7]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning.

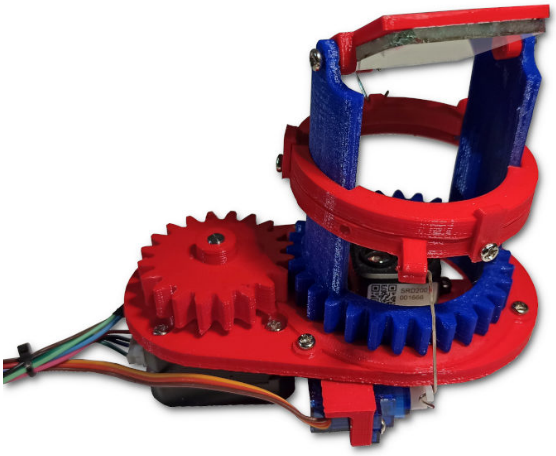


Figure 2.3: Implementation in project [7] using a 2D LiDAR mounted on a tilting platform

Pros

+ Faster data acquisition compared to solutions with single-point LiDAR measurements
+ Higher resolution given by the use of a 2D LiDAR sensor
+ This design could be potentially used for Simultaneous Localization and Mapping (SLAM) applications, given the higher speed

Cons

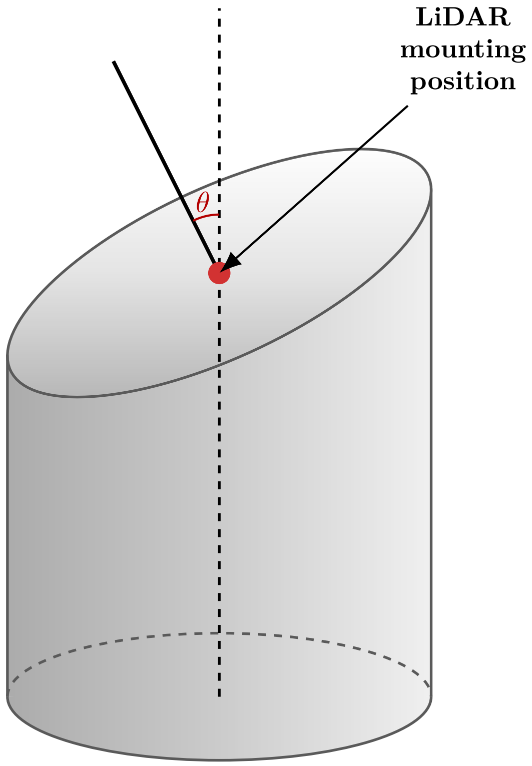
– More expensive than single-point LiDAR solutions

– Point distribution is non-uniform due to the tilting mechanism which produces denser point clouds on the axis of tilting (see MATLAB simulation below) [should this note be here?]

– Mechanical design is both more complex and more prone to vibrations produced by the LiDAR itself affecting measurement accuracy

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate 360°. This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high



resolution, while also eliminating some of the drawbacks of previous designs, namely:

- The platform will be quite rigid, minimizing vibrations
- Mechanical design is very simple, making it easy to build and troubleshoot
- The point cloud will be more uniformly distributed compared to the tilting platform design, as the rotation will ensure that a LiDAR won't have a preferred direction around which it collects denser data unless the slope is very steep (see MATLAB simulation below)[could this be phrased better?]

The main drawback of this design is the existence of "blind spots" in the point cloud, which are areas that the LiDAR cannot scan due to the tilt angle. However, this issue can be mitigated by carefully selecting the tilt angle based on the specific application in which the system will be used. (more on that in the next chapter) [system? :/]

2.2.1 Controlling the Platform

Since we want our platform to be precisely controlled, we need to choose an appropriate motor. Stepper motors are commonly used in applications requiring precise positioning, as they can be controlled to move in small, discrete steps. For this design, a NEMA 17 17HS8401 stepper motor is used to rotate the LiDAR platform.

Figure 2.4: NEMA 17 17HS8401 photo[1]

Table 2.1: Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor[1]

Motor Weight [Ω]	360
Holding Torque [N m]	0.5
Step Angle [°]	1.8
# of Steps per Revolution [-]	200
Rated Current [A]	1.68

For heavy load applications, it is important to consider the potential error sources of stepper motors, such as the stepper motor losing steps due to a heavy load or by driving the motor faster than its maximum speed [8, p. 13]. However, in our case, the load will be relatively light for this motor and the speed will be kept low, thus a feedback mechanism for correcting lost steps is not required.

[1] Photo and parameters were taken from `https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/`

2.2.2 Choosing the Tilt Angle

The tilt angle of the LiDAR platform is a crucial parameter that affects the size of the aforementioned blind spots in the point cloud. A larger tilt angle will result in smaller blind spots as could be seen from the simulations below.

The trade-off for a larger tilt angle, however, is a more spread-out point cloud in the horizontal direction if we rotate our platform in discrete steps [see point cloud simulations below]. Without any further solutions, we would have to choose a tilt angle that balances the size of blind spots and point cloud density.

Thankfully, there is a way to achieve a more complete coverage of the environment without sacrificing point cloud density.

2.2.3 Choosing a Gear Ratio for the Rotating Platform

In order to remove the need for a compromise between point cloud coverage and density while using a stepper motor to rotate the platform, we can introduce a gear mechanism into our design. Using gears with a specific gear ratios allows us to choose a tilt angle that minimizes blind spots while still maintaining a dense point cloud.

Chapter 3

Conclusion

Appendix A

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